



Acute Cough in Pediatrics

Diagnosing Acute Cough

Assessing the Patient

For acute cough, virtual history-taking should attempt to arrive at the most likely clinical diagnosis, when possible, and treat the underlying etiology. Most children with acute cough have viral upper respiratory tract infection (URI). In the absence of red flags, a period of watchful waiting and supportive care for 1-2 weeks is generally recommended for most children with non-specific acute cough.

Diagnoses other than viral URI should be considered in children who have cough that is more severe or prolonged than expected.

Virtual Physical Exam

- Assess general appearance and respiratory status via parental history or speaking directly to the child (e.g., "How does your child look?" "Is your child still active and playful?").
- Listen to child's breathing and/or cough over the phone or video.
- If necessary, ask the child to sing the ABCs to assess for obvious tachypnea, stridor, wheezing, shortness of breath, etc.

Red Flags (strongly consider an in-person evaluation)



Note. From Central Cyanosis [Photograph], by Azmi N.F.B. 2016, ResearchGate (https://www.researchgate.net/figure/Central-cyanosis-5_fig1_307191162).

- Prolonged fever (generally > 5 days or earlier, depending on age and other clinical circumstances)
- Shortness of breath
- Hemoptysis
- Concern for foreign body aspiration
- Very ill appearance
- Cyanosis
- Medically complex (e.g., recurrent pneumonias, immunodeficiency, Pertussis or TB exposure, etc.)

Treatment Considerations

Selecting the Most Effective Treatment

Nasal Saline Irrigation and/or Suctioning

- Primarily beneficial for nasal symptoms, but it may alleviate cough associated with nasal congestion/mucus/postnasal drip.

Honey

- A systematic review of 14 randomized trials concluded that honey is superior to placebo & OTC cough medications.
- Can be given straight or diluted in liquid (e.g., warm water, tea, juice)
- Do NOT recommend for children < 1 year of age (*Clostridium botulinum* risk)



Humidifier (Cool v. Warm?)

- AAP recommends use of a cool mist humidifier.
- Warm mist/vaporizers may pose a safety hazard due to steam and hot water.
- If using a warm mist humidifier, provide safety guidance (e.g., keep at least 3 feet away from child, avoid location where child can pull down, etc.).



Note. From Dextromethorphan in its syrup presentation [Photograph], by Rassnik, 2014, Wikimedia Commons (<https://commons.wikimedia.org/wiki/File:Dextrometorfano.jpg>).

OTC Homeopathic Remedies

- Currently, no homeopathic product meets FDA standard for safety and efficacy.

OTC Cough Suppressants/Expectorants Decongestants (e.g., Dextromethorphan, Guaifenesin, Phenylephrine)

- AAP recommends **avoiding** OTC cough and cold medications in children younger than 6 years of age.
- Even for children > 6 years of age, there is **no good evidence** that OTC cough and cold medications are effective, and they may cause adverse effects (confusion, dystonia, hallucinations, tachycardia, seizures, respiratory depression, nausea, vomiting, diarrhea).
- Beware of combination cough & cold medications that also contain an analgesic/antipyretic (risk of toxicity when combined with stand-alone acetaminophen or ibuprofen).

Antihistamines

- In randomized trials, antihistamines have not been shown to be effective for nasal symptoms or cough associated with viral URI.
- If history of potential atopy or allergic rhinitis with throat-clearing type of cough, may consider trial of oral antihistamine and/or intranasal corticosteroids.

Asthma Medications

- If patient is a known asthmatic, [treat accordingly](#).
- If **NOT a known asthmatic**, but has a persistent non-specific dry cough despite 2 weeks¹ of watchful waiting (especially in the presence of other atopic features—eczema, allergic rhinitis, FH of asthma), may consider empiric trial of asthma therapy (bronchodilator +/- low-dose inhaled corticosteroid).
- ¹ **Please Note:** Bronchodilators and inhaled corticosteroid would NOT be beneficial for routine cough/URI. Only consider if history is suggestive of potential asthma.
- If considering asthma: Recommend empiric trial of asthma medications over a pre-defined timeframe (2-4 weeks) and recommend follow-up reassessment.
- If cough is not improved over 2-4 weeks, then asthma therapy should be discontinued.
- If cough is improved over 2-4 weeks, then patient may have asthma (or possibly spontaneous resolution of cough).



Note. From Asthma in Children [Photograph], by Johns Hopkins Medicine, n.d., Johns Hopkins Medicine (<https://www.hopkinsmedicine.org/health/conditions-and-diseases/asthma/asthma-in-children>).

Oral Steroids

- Generally, only indicated for significant [acute asthma exacerbation](#) or [croup](#) in pediatrics.
- Avoid for otherwise non-specific cough/URI.
- If considering mild asthma for non-specific cough, consider asthma options noted above.

Nebulized Saline (normal saline or hypertonic saline)

- Not routinely recommended for cough or bronchiolitis, but more studies are needed.

Benzonatate (Tessalon perles)

- FDA approved for children > 10 years of age.
- May be considered, but not routinely recommended.
- Warning:** Accidental ingestion and potentially fatal overdose has been reported in children < 10 years of age.
- If prescribing, should warn caregivers NOT to give to young children in the household, and to safely store medications.
- Symptoms of toxicity: restlessness, tremors, convulsions, or coma.



Note. From 100mg generic Benzonatate Capsules [Photograph], by Revision17, 2014, Wikipedia (https://en.wikipedia.org/wiki/Benzonatate#/media/File:100mg_generic_Benzonatate_Capsules.jpg).

Codeine

- AAP recommends **AGAINST** codeine for pediatric patients.
- Do NOT prescribe in pediatrics.
- Adverse effects: somnolence, respiratory depression, and even death.

Antibiotics

- Not recommended for non-specific cough/viral URI.
- May consider for classic [sinusitis](#) history.
- If otherwise considering bacterial source of cough, should consider an in-person evaluation.

References

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